Maximal Dissociated Subsets of Finite Abelian Groups

Benjamin T. Shepard

Department of Mathematics, Gettysburg College
E-mail: shepbe01@gettysburg.edu

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Abstract

A subset A of an abelian group G is said to be dissociated if $0 \in G$ cannot be written as the signed sum of any of the elements of A. Furthermore, the dissociativity dimension of a set A, denoted $\dim A$, is the maximum size of a dissociated subset of A. We are interested in evaluating the function $\dim(G, m)$, which yields the minimum value of $\dim A$ when $A \subseteq G$ has size m. Here, we provide some basic values and bounds for this function, and evaluate it for certain elementary abelian p-groups. In particular, we obtain exact values when the lower bound is sharp for $p = 2$ and $p = 3$. We also disprove a conjecture made by B. Bajnok in 2018 regarding the exact value for cyclic groups.

1 Introduction

In this paper, we will refer to G as an arbitrary finite abelian group, and κ as its exponent. We will also let m be a positive integer. We begin with some definitions.

Definition 1. *Let G be a group. A subset $A = \{a_1, \ldots, a_m\} \subseteq G$ is dissociated if every possible equality of the form*

$$\sum_{i=1}^{m} \lambda_i a_i = 0,$$

where $\lambda_i \in \{-1, 0, 1\}$, implies that $\lambda_i = 0$ for all i. We usually denote the set of all possible sums as defined above by ΣA, and write $0 \notin \Sigma A$ if A is dissociated.

Dissociated subsets of a group have been investigated, however there is an interest in finding dissociated subsets of any subset of a group. For this, we define

Definition 2. *Let A be a subset of a group G. The quantity*

$$\dim A := \max\{|D| \mid D \subseteq A,\ D \text{ is dissociated}\}$$

is known as the dissociativity dimension of A in G.

Throughout this paper, we refer to $\dim A$ as simply the dimension of A. Finally, we will define the main function that we are interested in investigating.

Definition 3. *Let G be a group of order n, and $m \leq n$ be a positive integer. Define*

$$\dim(G, m) := \min\{\dim A \mid A \subseteq G,\ |A| = m\};$$

that is, the minimum dissociativity dimension of any subset of G of size m.

2 Previous results

We currently have the following:

Theorem 4 (Lev and Luster, 2011; [1]). *For any $A \subseteq G$, we have*

$$r_A \leq \dim A \leq \lfloor r_A \cdot \log_2 \kappa \rfloor$$

where r_A is the rank of the subgroup $\langle A \rangle$ generated by A.

Proposition 5 (Bajnok, 2018; [2]). *If G is of type $n_1, \ldots, n_r$ and order n, then*

$$\lfloor \log_2 n_1 \rfloor + \ldots + \lfloor \log_2 n_r \rfloor \leq \dim(G, n) \leq \lfloor \log_2 n \rfloor.$$

In particular, for all positive integers n, we have

$$\dim(\mathbb{Z}_n, n) = \lfloor \log_2 n \rfloor.$$

Proposition 6 (Bajnok, 2018; [2]). *For all groups G, we have $\dim(G, 1) = 0$, $\dim(G, 2) = 1$, and*

$$\dim(G, 3) = \begin{cases} 1 & \text{if } \kappa \geq 3; \\ 2 & \text{if } \kappa = 2. \end{cases}$$

3 Main results

We first evaluate the cases of $m = 4$ and $m = 5$ for all groups G in order to obtain an extension of Proposition 6.

Proposition 7. *For all G, we have*

$$\dim(G, 4) = 2$$

and

$$\dim(G, 5) = \begin{cases} 2 & \text{if } \kappa \geq 3; \\ 3 & \text{if } \kappa = 2. \end{cases}$$

Proof. When $\kappa = 2$, both claims follow from Theorem 14. Therefore, assume $\kappa \geq 3$. For $m = 4$, let

$$A = \{0, g_1, -g_1, g_2\}$$

such that $g_1, -g_1$, and g_2 are all nonzero and distinct. For $m = 5$, a non-boolean abelian group G of order at least 5 contains at least two elements $g_1, g_2 \in G$ such that $g_1, g_2 \neq 0, -g_1, -g_2$, so we can let

$$A_1 = A \cup \{-g_2\}$$

such that $A \cap \{-g_2\} = \emptyset$. Clearly both A and A_1 do not have a dissociated subset of size 3, so $\dim A \geq 2$ and $\dim A_1 \geq 2$. In both A and A_1, one can easily see that $0 \notin \Sigma\{g_1, g_2\}$, so $\dim A = \dim A_1 = 2$.

Since only subsets of a set of the form $\{0, g_1, -g_1\}$ have dimension less than 2, we have $\dim(G, 4) = \dim(G, 5) = 2$ as claimed. ■

We also introduce several bounds. This first proposition states that the dimension function is monotonically increasing over m, which is an important fact that will be used later.

Proposition 8. *If m_1 and m_2 are positive integers such that $m_1 \geq m_2$, then*

$$\dim(G, m_1) \geq \dim(G, m_2).$$

Proof. For simplicity, let $\delta_1 = \dim(G, m_1)$ and $\delta_2 = \dim(G, m_2)$. We must prove that every subset of G of size m_1 has dimension at least δ_2. Let $A \subseteq G$ be arbitrary

and of size m_1. Since $|A| \geq m_2$, A contains at least one subset B of size m_2. By definition of δ_2, every subset of G of size m_2 has dimension at least δ_2, so $\dim B \geq \delta_2$. Thus B contains a dissociated subset D of size δ_2, and $B \subseteq A$, so we have $D \subseteq A$. By definition of dimension, $\dim A \geq |D| = \delta_2$ as claimed. ■

Proposition 9. *For all G of order n, and all $m \leq n$, we have*

$$\dim(G, m) \leq \lfloor \log_2 n \rfloor.$$

Proof. By Proposition 5, $\dim(G, n) \leq \lfloor \log_2 n \rfloor$. Using Proposition 8, we obtain

$$\dim(G, m) \leq \dim(G, n) \leq \lfloor \log_2 m \rfloor$$

as claimed. ■

Proposition 10. *For all m, and all G for which $|\mathrm{Ord}(G, 2)| < m/2$, we have*

$$\dim(G, m) \leq \left\lfloor \frac{m}{2} \right\rfloor.$$

Proof. Let $k = \lfloor \frac{m}{2} \rfloor$. If m is even, then $m = 2k$; if m is odd, then $m = 2k+1$. Since less than $m/2$ elements in G have order 2, we may let

$$A = \{0\} \cup \bigcup_{i=1}^{k} \{a_i, -a_i\} \subset G$$

such that $a_i \neq -a_i \neq 0$ for all i and

$$\{a_i, -a_i\} \cap \{a_j, -a_j\} = \emptyset$$

for all $i \neq j$. If m is even, take A without $-a_k$. In either case, $|A| = m$.

Suppose that there exists a dissociated subset $D \subset A$ such that $|D| > k$. Since the smallest possible size of D is $k+1$, if we choose $k+1$ elements of A, we are forced to choose either 0 or both a_i and $-a_i$ for some $i \in \{1, \ldots, k\}$. In either case, $0 \in \Sigma D$, so we must have $|D| \leq k$, a contradiction. Therefore, $\dim A \leq k$, which directly implies that $\dim(G, m) \leq k$ as claimed. ■

Note that these bounds are not generally sharp. For large m, Proposition 9 is better, but note that it is in terms of n, which loosens the bound.

In the next theorem, we introduce a lower bound for all G and m. First, we have a useful fact that assists with this theorem.

Lemma 11. *If G has invariant factorization*

$$G \cong \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_r},$$

then the group

$$H = \mathbb{Z}_{n_{r-k+1}} \times \cdots \times \mathbb{Z}_{n_r}$$

is the largest subgroup of G with rank k.

We will not provide a proof for this lemma here. However, it allows us to obtain the following result:

Theorem 12. *Suppose that G has invariant factorization*

$$G \cong \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_r}.$$

Then for all m we have

$$\dim(G, m) \geq k$$

for the unique k such that

$$\prod_{i=r}^{r-k+2} n_i < m \leq \prod_{i=r}^{r-k+1} n_i.$$

Proof. Let $A \subseteq G$ have size m. Since

$$\prod_{i=r}^{r-k+2} n_i < m,$$

the size of A is strictly greater than that of the subgroup

$$\mathbb{Z}_{n_{r-k+2}} \times \cdots \times \mathbb{Z}_{n_r}.$$

By Lemma 11, this is the largest subgroup of G with rank $k-1$. Therefore, $\langle A \rangle$ has rank $r_A \geq k$. Now from Theorem 4 we have $\dim A \geq k$, and the claim follows. ■

From this we immediately obtain

Corollary 13. *For all positive integers k and r, we have*

$$\dim(\mathbb{Z}_k^r, m) \geq \lceil \log_k m \rceil.$$

Proof. Since $k^{i-1} < m \leq k^i$ for some $i \in \{1, \ldots, r\}$, we have $i \geq \lceil \log_k m \rceil$. The claim now follows from Theorem 12. ■

It should be noted that for $k > 3$, equality does not generally hold. For example, it is an immediate consequence from Theorem 4 that

$$\dim(\mathbb{Z}_k^r, k^r) \geq kr$$

for all r and $k \geq 4$, which is larger than the lower bound of r given here. We do not currently know much about other values of m.

It turns out that the bound in Corollary 13 is sharp for certain elementary abelian p-groups; in particular, $p = 2$ and $p = 3$. We obtain both

Theorem 14. *For all positive integers r and $m \leq 2^r$, we have*

$$\dim(\mathbb{Z}_2^r, m) = \lceil \log_2 m \rceil.$$

and

Theorem 15. *For all positive integers r and $m \leq 3^r$, we have*

$$\dim(\mathbb{Z}_3^r, m) = \lceil \log_3 m \rceil.$$

The proofs of Theorems 14 and 15 are similar, so here we prove both using a general construction.

Proof. Let $p \in \{2, 3\}$. Since p is prime, $\mathbb{Z}_p$ is a field and $\mathbb{Z}_p^r$ forms a vector space over $\mathbb{Z}_p$. Recall that for any $A \subseteq \mathbb{Z}_p^r$, the coefficients of sums in ΣA are given by $\lambda_i = \{0, 1, -1\}$. The scalars in $\mathbb{Z}_3^r$ are exactly λ_i. Also, since each element of $\mathbb{Z}_2^r$ is its own inverse, for $p = 2$ we only need to consider $\lambda_i = \{0, 1\}$, which are exactly the scalars in $\mathbb{Z}_2^r$. Therefore, for any $X \subseteq \mathbb{Z}_p^r$, we have an equivalence: X is dissociated if and only if the vectors in X are linearly independent.

We now use Proposition 8 to narrow down the cases; it suffices to prove that $\dim(\mathbb{Z}_p^r, p^k) = k$ and $\dim(\mathbb{Z}_p^r, p^k + 1) = k + 1$ for all positive integers $k \leq r$.

First, let $m = p^k$. From Theorem 13, we have $\dim(\mathbb{Z}_p^r, p^k) \geq k$, so it suffices to show the existence of subset of $\mathbb{Z}_p^r$ of size p^k and dimension k. Let

$$A = \{0\}^{r-k} \times \mathbb{Z}_p^k.$$

It is clear to see that $\dim A = \dim \mathbb{Z}_p^k$. Now, it is a well known fact that if a vector space V is spanned by k vectors, and $l > k$, then any set of l vectors in V is linearly dependent. Since the standard basis for $\mathbb{Z}_p^k$,

$$\mathcal{B} = \{(e_1, \ldots, e_k) \mid e_i = 1 \text{ for some } i \in \{1, \ldots, k\},\ e_j = 0 \text{ for } j \neq i\},$$

spans $\mathbb{Z}_p^k$ and has size k, any subset of $\mathbb{Z}_p^k$ must have size at most k to be dissociated. It is clear that $0 \notin \Sigma\mathcal{B}$ since $\mathcal{B}$ is linearly independent, so $\dim \mathbb{Z}_p^k = k$. Therefore, we obtain $\dim A = k$ as desired.

Now let $m = p^k + 1$. Theorem 13 yields $\dim(\mathbb{Z}_p^r, p^k + 1) \geq k + 1$, so it again suffices to show existence of a subset of $\mathbb{Z}_p^r$ of size $p^k + 1$ with dimension $k + 1$. Write

$$A_0 = A \cup \mathcal{B}_0$$

where

$$\mathcal{B}_0 = \{1\} \times \{0\}^{r-1}.$$

Since $\mathcal{B} \cap \mathcal{B}_0 = \emptyset$, the set

$$(\{0\}^{r-k} \times \mathcal{B}) \cup \mathcal{B}_0 \subset A_0$$

has size $k + 1$ and is clearly dissociated.

Now, suppose that there exists a dissociated subset $D \subset A_0$ with $|D| > k + 1$. Since $\dim A = k$, D cannot contain more than k elements of A. But since we have $|A_0| = k + 1$, it is impossible to put two more elements of A_0 into D. Hence $|D| \leq k + 1$, a contradiction. From this, we obtain $\dim A_0 = k + 1$ as desired. ■

This proof does not work for $p \geq 5$; even though $\mathbb{Z}_p^r$ is a vector space for all values of p, the only groups where equivalence holds between dissociativity and linear independence are $\mathbb{Z}_2^r$ and $\mathbb{Z}_3^r$.

Using Proposition 5, we can easily get equality for a few related groups when m is of maximum size.

Proposition 16. *For all positive integers r and k, we have*

$$\dim(\mathbb{Z}_2^r \times \mathbb{Z}_{2k}, 2^{r+1}k) = \lfloor \log_2 2^{r+1}k \rfloor.$$

Proof. From Proposition 5, we have

$$r + \lfloor \log_2 2k \rfloor \leq \dim(\mathbb{Z}_2^r \times \mathbb{Z}_{2k}, k2^{r+1}) \leq \lfloor \log_2 k2^{r+1} \rfloor.$$

Since clearly

$$r + \lfloor \log_2 2k \rfloor = r + 1 + \lfloor \log_2 k \rfloor$$

and

$$\lfloor \log_2 k2^{r+1} \rfloor = \lfloor \log_2 2^r \cdot 2 \rfloor + \lfloor \log_2 k \rfloor = r + 1 + \lfloor \log_2 k \rfloor,$$

the bounds are equal and equality holds. ■

We now move on to cyclic groups $\mathbb{Z}_n$. In [2], B. Bajnok conjectured that for all positive integers n and $m \leq n$, the following equality holds:

$$\dim(\mathbb{Z}_n, m) = \lfloor \log_2 m \rfloor.$$

Using the code provided by Francis in [3], we have the following table of values that disagree with this conjecture:

G	m	$\dim(G, m)$	$\lfloor \log_2 m \rfloor$
$\mathbb{Z}_{17}$	$14, 15$	4	3
$\mathbb{Z}_{19}$	$14, 15$	4	3
$\mathbb{Z}_{22}$	15	4	3
$\mathbb{Z}_{23}$	$14, 15$	4	3
$\mathbb{Z}_{26}$	15	4	3
$\mathbb{Z}_{28}$	15	4	3
$\mathbb{Z}_{29}$	$14, 15$	4	3

Table 1: Values larger than the conjecture.

All other groups below $n = 30$ agree exactly for all values of m. In light of these results, I believe that the lower bound still holds.

I propose the following more modest conjecture:

Conjecture 17. *For all positive integers n and $m \leq n$, we have*

$$\dim(\mathbb{Z}_n, m) \geq \lfloor \log_2 m \rfloor.$$

I also believe that powers of two agree with Bajnok's conjecture.

Conjecture 18. *For all positive integers n and $m \leq n$, we have*

$$\dim(\mathbb{Z}_n, 2^k) = k.$$

It may also be more feasible to consider specific cyclic groups. For instance, the case when n is a power of two may be of particular interest.

4 Future work

Future projects should explore the validity of Conjectures 17 and 18. Additionally, since the upper bounds of Proposition 10 and 9 are not always sharp, future work should look into the possibility of improving these bounds.

Furthermore, since Corollary 13 is only sharp for $k = 2$ and $k = 3$, future work should attempt to improve the lower bound for $k \geq 4$, and possibly other elementary abelian p-groups for $p \geq 5$.

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